DE[X] DOUBLE END (DE) High Output Lamp Fixture
END TO END INSTALLATION INSTRUCTION



CAUTION! Never Expose Eyes or Skin to UV-C - Read all Materials before Starting

SAFETY CONSIDERATIONS

Improper installation, adjustment, alteration, service, maintenance, or use can cause fire, electrical shock, or other conditions which may cause personal injury or property damage. Consult a qualified installer, service agency, or your supplier for information or assistance. A qualified installer or agency must use factory kits or accessories when installing this product. Refer to the individual instructions packaged with kits or accessories when installing them. Follow all safety codes, wear safety glasses, and work gloves. Read all instructions thoroughly and follow any warnings or cautions attached to any accessed area. Consult local building codes and the National Electrical Code (NEC) for special requirements.



Understand the signal words **DANGER, WARNING or CAUTION**. These words are universally used for overall safety. **DANGER** identifies the most serious hazards, which will result in severe personal injury or death. **WARNING** signifies hazards, which could result in personal injury or death. **CAUTION** is used to identify unsafe practices, which would result in minor personal injury or product and property damage.



WARNING: Before installing or servicing this unit, turn off or disconnect all power. There may be more than one (1) switch.

CAUTION-UV EXPOSURE PRECAUTIONS:

1. Doors, covers, lamp mounting brackets, access panels, and removable components that give direct access to Ultraviolet (UV) lamp systems shall be equipped with an interlocking mechanism that removes power from the Ultraviolet (UV) lamp system.
2. There should be no openings, holes, or joints that allow UV light leakage.
3. Drain pans shall be metallic.
4. Some UV installations may require protective barriers (not supplied) to protect users, service personnel and non-metallic parts from UV energy.

BARRIERS (if required, not supplied):

a. Barriers should be constructed of a material that will not be affected by UV energy (e.g. metal)
b. Non-Metallic materials may be protected by directly attached barriers such as aluminum coated tape.
c. Both the unit inlet and outlet should be considered as possible Ultraviolet (UV) radiation paths when determining barrier locations.
d. The unit filters are not considered Ultraviolet (UV) radiation barriers.
e. *Removable barriers shall be marked with the words "Caution" and the following or equivalent:*

"UV LIGHT SOURCE - KEEP PROTECTIVE BARRIER IN PLACE"



CAUTION-Material Exposure / Degradation:

1. Any non-metallic parts exposed to UV shall be shielded.

Note: Metallic barriers, and aluminum tape are appropriate means of shielding. Non-metallic parts include, but are not limited to conductors, connectors, enclosures,

housings, drain pans, support pieces, gaskets, and filters.

2. Protective materials can include barriers (referenced above), conduit, aluminum tape, etc. Please reference ASHRAE® 2016-Systems and Equipment handbook for additional information[1].
3. Inorganic materials such as glass, glass fibers, and metal are not affected by UV-C exposure[123].
4. Shield all organic material components within 5 ft **(**1.5 m) of the UV lamp[1].

GENERAL:

1. DE Fixtures can be used within an air handler to bathe surfaces, the air, or both with UV-C energy. For surfaces, locate fixtures at known growth areas such as coils, drain pans, and/or around humidifiers, etc.
2. Before starting, check for power availability, and ensure the component mounting locations to the on/off toggle switch, door safety switch, UV-C Hour Meter and radiometer (if used).

NOTE: One light fixture may power up to a maximum of 5 additional light fixtures with the wire plugs and wire receptacles included in "DE ELECTRICAL CONNECTION KIT" P/N 52000045. The plug / receptacle set P/N is 90004406.

NOTE: The ends of each light fixture employ a conduit opening. On the 1st UV light fixture of the end to installation, the electrician must install his field wiring leads into one of the conduit openings. The other end of the 1st light fixture must be installed with the supplied receptable. For subsequent light fixtures, the male plug (with protruding pins) should only be installed in the conduit opening on the non-power side of the fixture to be plugged into the power side of the female receptacle of the prior light fixture. A maximum of 5 subsequent light fixtures (up to a total of 6) may be powered from one light fixture connected to the field installed power supply.

INCLUDED:

- **DE Plug-n-Play (PnP) Fixture**
- **UV-C Lamp (Typ) (Note: Fixture also sold without lamp)**
- **DE installation Kit:**
  - **Installation Operation Manual (IOM)**
  - **Door Warning Label**
  - **Product Specification Label**
  - **Self-Drilling Mounting Screws (x4)**
  - **Fixture Top Screws (x2)**
  - **Dome Plug (x1)**
- **Installation Process (structural surfaces only):**



A. Consult all applicable codes and the NEC before installation. Check fixture label(s) for the correct power requirements and supply power from a suitable, protected (fused), and grounded power source.



CAUTION-DANGER: **Using other than the labeled voltages will void product warranty and may damage the entire system. It is recommended that lamps be operated continuously to avoid mold and bacteria growth, especially when the fan is off.**

[1] ASHRAE-2016. UV-C Photodegradation of Materials. Systems and Equipment Handbook. 17.5-17.6

[2] Kaufman, R.E. 2017. Study the HVAC System Photodegradation Caused by the Low Level UVC Light Irradiance Used for Coil Maintenance and Air Stream Disinfection. ASHRAE Research Project 1724, Final Report

[3] Kauffman, R. 2011. Study the degradation of typical HVAC materials, filters and components irradiated by UVC energy. ASHRAE Research Project RP-1509, Final Report.

B. All UV-C system access must be power-interlocked from the source to turn lamp(s) off when the UV-C system is accessed. Access points should have signs in appropriate languages alerting maintenance personnel to the possible hazard of looking at or exposing skin to UV-C energy (English label provided). Lamp access may be equipped with glass or Lexan® windows to safely view their operation. Ordinary glass or Lexan® will typically block UV-C energy. It's recommended that any glass be tested for UV-C transmission.

C. Determine the most desirable location for the intended use and install reflective material (if used).

D. A fixture may be used as a template to mark and drill mounting holes for your application. Install fixture(s) and wires in accordance with all applicable national, state and local codes. (see Installation Suggestions).

E. Before installation, clean the lamps with isopropyl alcohol and a lint free cloth (recommended). Grasp the non-glass ends (thumb and index fingers) and carefully push lamp pins into the socket pin slots until the lamp is seated. Rotate until one distinct click is heard. **Note: Lamp must click to ensure the lamp is safely and properly seated.**

Installation Suggestions - DE Products

Suggestions are intended for qualified installers only. Refer to layout drawings (if supplied) when using the following suggestions:

1. Support material (Not Supplied/ By others)

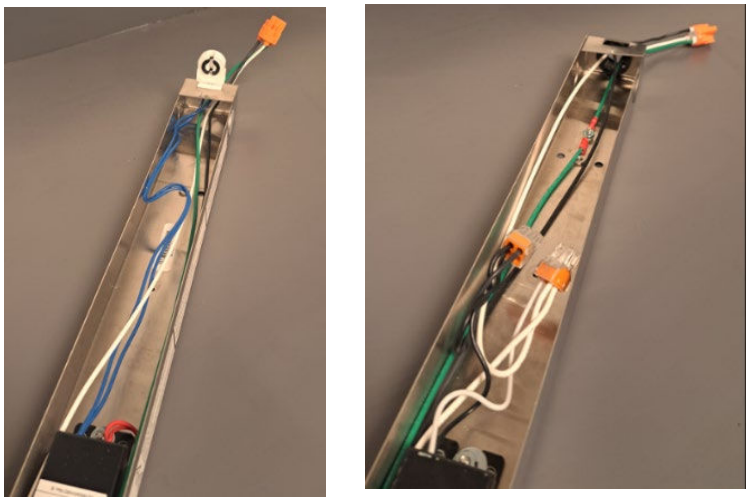
☐ Use the appropriate gauged, pre-formed galvanized, stainless, or aluminum angle for each row's horizontal support. Uni-Strut®, Kindorf® or an equivalent may also be used and cut to size in the field. These same materials may serve as vertical support when required.

NOTE: When more than one angle is used to span the fixture width or height, overlap the angles at the fixtures mounting holes to provide a stronger structural joint.

2. Install row(s) as follows:

☐ From the layout drawing example located in Page 7, note all the corresponding dimensions, (e.g. from the coil to the UV-C fixture(s), from the bottom of the plenum to the first row, and between rows (row to row)). Mount the vertical supports, then proceed mounting the horizontal supports using the layout drawing as a reference.

☐ Unpack fixtures and remove lamp and fixture cover and set aside (Pic. 1).



Pic 1: Unpacked fixtures with removed lamp and fixture cover.

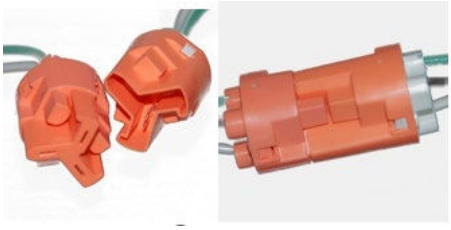


Pic 2: Remove Lamp and the Cover. Set to the aside.

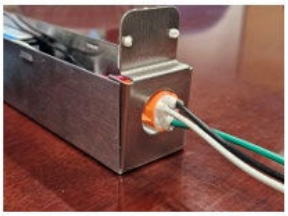
☐ Position the correct number and size of fixtures for each row, end to end for horizontal support, making sure that plug-n-play (PnP) plugs (male to female) mate properly (Pic. 2).

NOTE: MAKE SURE THAT THE WIRE COLORS MATCH BEFORE CONNECTORS ARE MATED.

☐ Plug each fixture into the next (Pic. 3) and tuck the mated plugs into the fixture *without* the plastic brushing (Pic. 4).

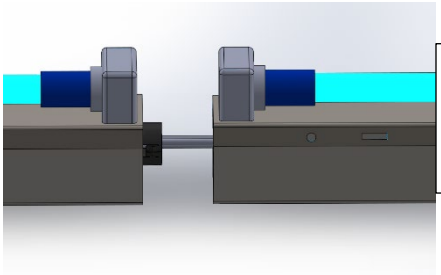


Pic 3: Properly mate the plug and play connectors.

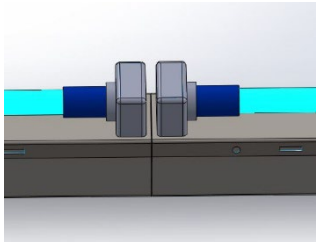


Pic 4: Plug tucked into the fixture.

☐ Push the fixtures together until the bushing slides (“clicks”) all the way into the next fixture (Pic. 5 & 6).



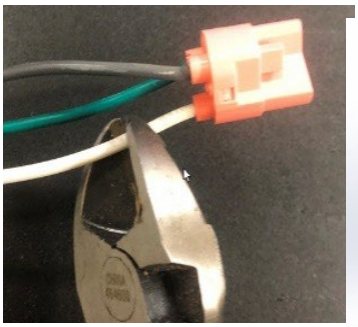
Pic 5: Push fixtures together until bushing clicks together. (Note: Fixture top will need to be removed to secure to support structure.)



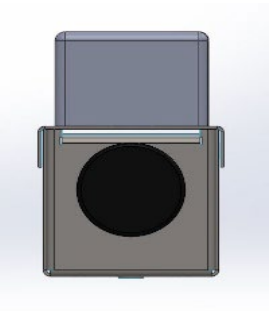
Pic. 6: Fixtures together.

☐ Repeat this process with all the fixtures necessary to complete the row.
☐ With the fixtures mounted on the horizontal support, position them to allow for the completion of the wiring at one end. When a longer fixture is part of a row, it is recommended for it be placed last, where the wiring will end.
☐ Cut off PnP plug at the wire-to-power end to accommodate wiring to power (Pic. 7).Rephrase
☐ Tuck the PnP plug on the unused end into its fixture and seal the fixture’s end by installing a “dome plug” (supplied) (Pic. 8).

☐ Use the self-drilling screws (supplied) to fasten the fixtures to the horizontal support as needed.
☐ Continue this procedure until “all” rows (if more than one) are completed.
☐ Reinstall the removed fixture covers. Use the supplied screws to fasten covers.
☐ Use the “dome plugs” to cover any open ½” knockouts remaining (Pic 8).



Pic. 7: Cut off PnP Plug.

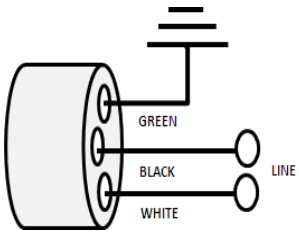


Pic. 8: Dome Plug placed at the end of the fixture.

☐ **REMOVE THE REFLECTOR PROTECTIVE COVER BEFORE INSTALLING LAMPS TO THE FIXTURE(S).**

3. Wiring fixtures and terminating.

1. Using the appropriate electrical components (not supplied) per the NEC, State and local codes, bring the appropriate power to the "finish wiring" end of fixtures via conduit to finish all the connections.
2. Use available metal or aluminum tape to cover and protect any UV-C susceptible materials (see material degradation above).



Dwg 1: Wiring Diagram

NOTE: Maximum number of fixtures that can be PnP wired together in rows cannot exceed 8A. Refer to ballast label on fixture for specific amps relating to each fixture at different voltages or Power Consumption chart below

4. Labeling Instructions

☐ Apply "System Lamp Watt Specification Label" to the exterior of the location where fixtures are installed. Complete label by marking the quantity of the specific fixture(s) installed in the unit and the total lamp watts.

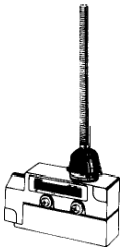
☐ If more than one fixture is connected to the power line, affix "number of fixture" label included in the DE ELECTRICAL CONNECTION KIT on the fixture.

5. Power considerations

☐ Read the fixture label power requirements – sum the power requirements, wire, and fuse accordingly.

☐ Run power-in wiring from a suitable, protected and grounded power source and in accordance with NEC, State and local codes. Power should be switched from outside the plenum and "**must"** run through a properly rated, SPST interlock door switch (Pic. 9), at each access.

☐ Apply supplied Door Warning Label(s) (Pic. 10) on all access doors/ports/covers/panels.


Pic. 9: SPST Interlock Door Safety Switch.



Pic 10: Warning Label (supplied).

6. Lamps

☐ Wiping all lamps down with isopropyl alcohol and a lint free cloth before using is recommended (this step is not necessary for the EncapsuLamp™ – FEP coated/sleeved lamp).

☐ Lamps should be replaced annually (9000 hrs).

ELECTRICAL RATING: 3 Amps, 120V Maximum / 1.5 Amps, 240V Maximum

Power Consumption:

MODEL #	Part Number	Watts	Amps at 120V	Max °F	Lamp Watts
GTD 22 VO	20000200	68	0.58	150°F	40
GTD 34 VO	20000400	101	0.82	150°F	69
GTD 40 VO	20000500	113	0.91	150°F	92

Compliance: Underwriter Laboratories (UL) ABQK, Accessories, Air Duct Mounted. Tested in accordance with ANSI/UL 1995.

The health aspects associated with the use of this product and its ability to aid in the disinfection of environmental air have not been investigated by UL. Air duct mounted accessories with respect to electrical shock, fire and casualty hazards only.

NOTE: Grounding should be line-to-neutral (see Dwg. 1 for wiring diagram).

WARNING: Eye damage may result from directly viewing the light produced by these lamps. To reduce the risk of exposure to UV radiation, take UV-radiation protective measures for personnel during servicing.

WARNING: This fixture is designed for use with germicidal UV-C lamps and must be installed in compliance with competent technical directions so that the user's eyes and bare skin will not be subjected to injurious rays.

WARNING: There shall not be any supply or return air openings or any other openings that are in the direct line of sight of the UV bulb.

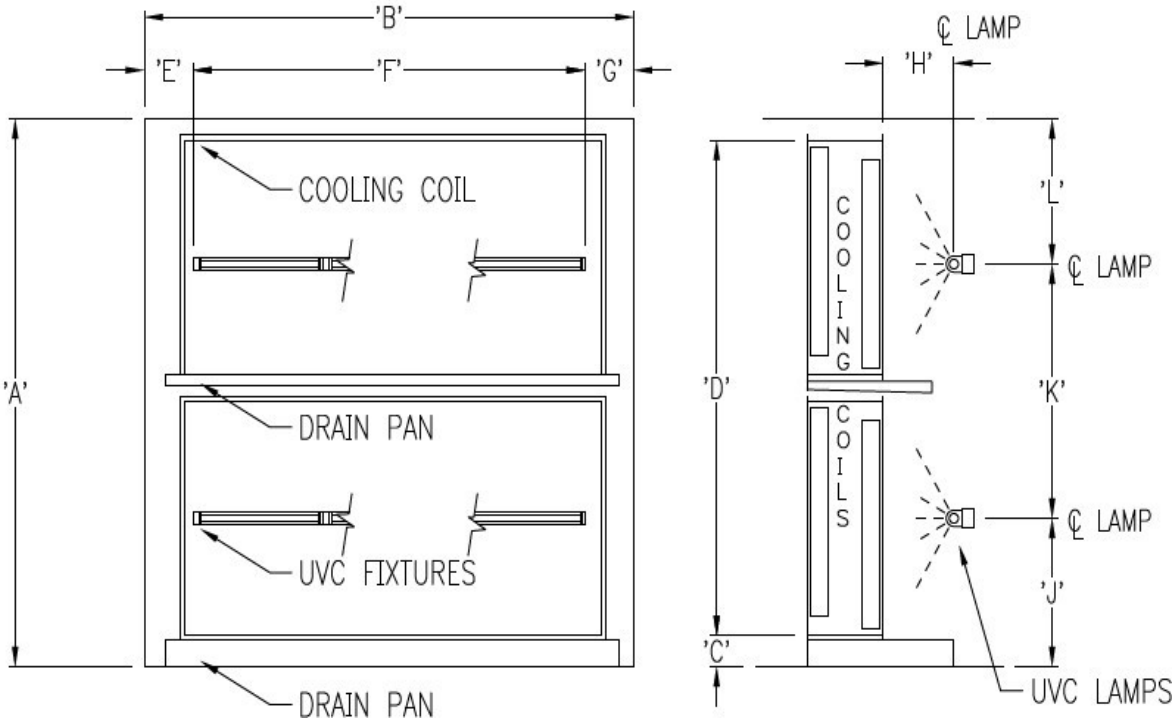
WARNING: There shall not be any openings in the duct that would emit light.

WARNING: UV Light Hazard. To Prevent exposure to ultraviolet light, be sure the ultraviolet air treatment system is disconnected before servicing any part of the HVAC system or removing any access panel or the equivalent.

CAUTION: Personal Injury Hazard. Power supply can cause electrical shock. Disconnect power supply before servicing or beginning installation.

CAUTION: Risk of exposure to excessive ultraviolet (UV) radiation - Do not operate without complete lamp enclosure in place or if sight lens is damaged.

LAYOUT DRAWING EXAMPLE



MAINTENANCE:

The intensity of the ultraviolet energy emitted from the UV lamps is dependent on the cleanliness and lamp age. The surface of the lamp should be kept as clean as possible for optimum intensity. Depending on the filtration level of the HVAC system and the general hygiene of the building, periodic cleaning may be necessary. Before attempting any maintenance procedures, always follow all warnings and cautions as detailed in this maintenance section.

CLEANING LAMPS

Note: If lamps are found to be broken, see the proper warning and cautions below regarding broken lamps and hazardous vapors.

1. Disconnect all electrical power to the unit and the UV lamps.
2. Wearing soft cloth gloves and safety glasses, grasp the lamp with fingers and twist lamp until it can be removed from tombstone.
3. Wipe down each lamp with a clean cloth and isopropyl alcohol. Avoid touching the lamp glass with hands as skin oils can accelerate lamp degradation. (If lamps are coated with Teflon they can be touched with bare hands).
4. Apply dielectric grease to the 4 pins. Place the lamp back in tombstone and twist the lamp until it clicks into place.

REPLACING THE LAMPS ANNUALLY

Ultraviolet lamps should be replaced annually if operated continuously or after 9,000 hours of use if operated intermittently. Replacement lamps must be the specific size and wattage as originally supplied by the factory.

Note: Although the lamps may continue to generate a characteristic blue glow beyond 9,000 operating hours, the ultraviolet energy emitted by the bulbs degrades over time and will no longer provide the intended benefit.

LAMP DISPOSAL:

UV lamps should be treated the same as other mercury-containing devices, such as fluorescent bulbs, according to local regulations. Most lamps must be treated as hazardous waste and cannot be discarded with regular waste. Low-mercury bulbs often can be discarded as regular waste; however, some states and local jurisdictions classify these lamps as hazardous waste. The U.S. EPA's universal waste regulations allow users to treat mercury lamps as regular waste for the purpose of transporting to a recycling facility. The National Electrical Manufacturers Association (NEMA) maintains a list of companies claiming to recycle or handle used mercury lamps at www.lamprecycle.org.

LIMITED EQUIPMENT WARRANTY

FOR DETAILED INFORMATION, VISIT https://uvresources/warranty.

The warranty (https://uvresources/warranty) supersedes and replaces any warranty statements orally made by a Salesperson, Distributor or Dealer or contained in the written instructions or other Brochures or informational documents in relation to this product. This warranty gives you specific legal rights, and you may also have other rights which vary from State to State.

Specifications subject to change without notice - © 2025, all rights reserved.